List of Technologies developed (domain-wise) by C-DAC

1. Domain: High Performance Computing (HPC)

Sr. No.	Parameters	Brief details
1.	**Brief details of Technology Developed**	Trinetra network is Indigenously developed High Performance Network solution intended as backbone of large scale HPC systems.
	Name of the Project under which technology is developed	National Supercomputing Mission (NSM)
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	9
	Brief Write-up on the Technology Developed	Trinetra HPC network is high performance, indigenously developed HPC network designed to provide low latency, high bandwidth network fabric for Supercomputing systems. Trinetra-B, the latest offering is based on PCI-e Gen3 (x16) host interface and multiple ports (10) based on 200Gbps physical link layer technologies. It is available as a standard PCI-e network interface card (NIC) footprint, and can be integrated in any standards-based server (Intel/AMD/ARM, etc). A standards based lightweight network stack allows standard MPI based applications to run seamlessly over Trinetra. A unique design feature of Trinetra is integrated packet switching which eliminates need of any external switching hardware thus providing power and cost savings for large scale systems.
	Details of Contact Person, C-DAC Centre along with Mobile No. and e-mail ID	Atul Bodas, C-DAC Pune atulb@cdac.in 9423567977
2.	**Brief details of Technology Developed**	Highly dense form factor RUDRA server based on Intel Xeon processors and Nvidia H200 GPUs
	Name of the Project under which technology is developed	National Supercomputing Mission (NSM)
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	9
	Brief Write-up on the Technology Developed	Rudra-SPX is based on Intel's Eagle Stream Platform. It is designed with Intel's fourth-generation Xeon Scalable Processor (SP) Sapphire Rapids (SPR) with Intel Chipset Emmitsburg. It supports next-gen industry-standard technologies such as PCIe 5.0, HBM & DDR5 memory speeds of up to 5600MT. It is compatible with Open Compute Platform (OCP) 48V DC for Cloud/Data center and HPC market. For cooling, it has the option of both Liquid (better energy efficiency) and Air cooling. The server is designed with interfaces required for HPC system and in small form factor. For further details, please refer https://www.cdac.in/index.aspx?id=product_details&productId=PARAMRudra

Sr. No.	Parameters	Brief details
	Details of Contact Person, C-DAC Centre along with Mobile No. and e-mail ID	Chinmay Bahulekar, C-DAC Pune chinmayb@cdac.in 9421185762
3.	**Brief details of Technology Developed**	Build on the success of the Traditional PARAM SHAVAK, Centre for Development of Advanced Computing (C-DAC) introduced PARAM SHAVAK 2.0, the next generation of ready-to-use, table-top HPC systems. Powered by the indigenous Rudra server and Bharat Operating System Solution (BOSS), PARAM SHAVAK 2.0 comes pre-loaded with essential software and customizable applications specifically tailored for scientific and engineering domains. This all-in-one solution aims to empower research organizations and academic institutions as they embark on their journey towards adopting HPC and fostering new skill sets. PARAM SHAVAK 2.0 offers three distinct variants to cater to diverse needs: • PARAM Shavak HPC: Ideal for general-purpose high-performance computing tasks. • PARAM Shavak DL GPU: Equipped with powerful GPUs, specifically designed for deep learning and other AI workloads. • PARAM Shavak VR: Optimized for virtual reality and immersive computing applications
	Name of the Project under which technology is developed	National Supercomputing Mission (NSM)
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 9
	Brief Write-up on the Technology Developed	PARAM Shavak **"A table top HPC solution"** powered by the indigenous Rudra Server, holds immense commercial potential, offering vast opportunities across various industries, startups, academia, and applications. PARAM Shavak is regarded as the gateway to large High-Performance Computing (HPC) clusters, as it effectively emulates the functionality of such clusters through its integration of the same software stack utilized in HPC clusters. With its immense computational power, compact design, power efficient, and low maintenance, PARAM Shavak caters to a diverse user base, including small and medium-sized businesses, research institutions, academia, and individual professionals. Furthermore, the commercialization of desktop supercomputers encourages innovation in software development, allowing for the creation of specialized applications tailored to specific industries and use cases. This ecosystem of software tools enhances the value proposition of desktop supercomputers, empowering users to tackle real-world problems effectively. As technology progresses and desktop supercomputers become more accessible and affordable, their commercial impact is poised to grow substantially in the coming years. Collaborating with local distributors in national and international markets, C-DAC ensures a successful market entry for PARAM Shavak, complemented by targeted marketing campaigns and strategic partnerships. Engaging key stakeholders across government agencies, research institutions, and academia, C-DAC orchestrates awareness meetings, workshops, and theme-based competitions to foster a culture of innovation and collaboration within the

Sr. No.	Parameters	Brief details
		academic community, further solidifying PARAM Shavak's position as a leading solution for computational needs.
	Details of Contact Person, C-DAC Centre along with Mobile No. and e-mail ID	Prashant Dinde, C-DAC Pune Email: prashantd@cdac.in Contact Number: 9822650512
4.	**Brief details of Technology Developed**	ICECloud: Integrated Computing Environment along with pipelines for genomics analysis and drug discovery as services
	Name of the Project under which technology is developed	Integrated Computing Environment (ICE) for Next Generation Biology: Cloud based HPC & Big Data facility-BIC- NSM-NPGDD
	Funding Agency	MeitY and DBT
	Technology Readiness Level (TRL)	TRL: 7
	Brief Write-up on the Technology Developed	ICECloud is a scalable cloud-based multi-omics platform for addressing translational research and precision health challenges. ICECloud platform is a modular, microservices-based system built using containerized workflows (Docker/Singularity) orchestrated via Kubernetes and workflow managers such as Nextflow, enabling scalable and reproducible execution across HPC and cloud environments. It supports end-to-end multi-omics analyses, including genomics, transcriptomics, proteomics, and metabolomics, seamlessly integrated with advanced structural biology pipelines for protein structure prediction, variant impact analysis, protein–ligand docking, macromolecular complex modeling, visualization, and cryo-EM data processing. Dedicated integration modules map omics-derived signals to protein structures, enable network and pathway analysis with structural overlays, and incorporate AI/ML methods, such as graph neural networks, to uncover structure-informed multi-omics correlations. The platform provides rich, interactive visualization interfaces through Jupyter Notebooks, R/Shiny dashboards, and Plotly, while ensuring robust security and governance via role-based access control, data encryption, and comprehensive audit logging, making it suitable for collaborative, large-scale biomedical research and translational applications **URL: https://icecloud.in**
	Details of Contact Person, C-DAC Centre along with Mobile No. and e-mail ID	Dr. Uddhavesh Sonavane, C-DAC Pune, uddhaveshs@cdac.in 86050 88425